

Review Article

LEO Mega Constellations: Review of Development, Impact, Surveillance, and Governance

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The rapid development of Low Earth Orbit (LEO) mega constellations has significantly contributed to several aspects of human scientific progress, such as communication, navigation, and remote sensing. However, unrestrained deployment of constellations has also strained orbital resources and increased spacecraft congestion in LEO, which seriously affects the safety of in-orbit operations of many space assets. For the long-term and sustainable development of space activities in LEO regions, space environment stability must be maintained using more rational surveillance and governance mechanisms. This review contributes to the research gap and facilitates the development of LEO mega constellations. First, the current development of typical LEO mega constellations is reviewed, followed by the analysis of the impact of LEO mega constellations in terms of astronomical observation, spacecraft safety in orbit, and space environment evolution. Then, two main solutions to conduct the challenges raised by LEO mega constellations are elaborated: one is to ensure the safety operation of spacecraft using space surveillance infrastructures and space situational awareness technologies, and the other is to accelerate the deorbit of constellation satellites at the end of life based on postmission disposal and active removal methods. Finally, the future development and potential research directions of LEO mega constellations are prospected.

1. Introduction

Since the successful launch of the first artificial Earth satellite, Sputnik-1, in 1957, space activities have presented significant value to society in terms of scientific and technological innovation, economic development, and national security. However, the near-Earth space environment has become congested with many space objects in the past several decades, as shown in Figure 1. The Low Earth Orbit (LEO) is the most congested region in near-Earth space that contains approximately 80% of space objects in only 0.3% of the space below the geosynchronous orbit (GEO) altitude [1].

With the proposal and deployment of many LEO mega constellations in recent years, the number of LEO objects has increased significantly. LEO mega constellations possess the characteristics of many satellites, a wide range of distribution, and a vast scale. These characteristics provide a broad range of applications in the fields of communication, remote sensing, and navigation, as well as bring revolutionary changes to the development of the global space industry. The United States, the United Kingdom, China, Canada,

Japan, Russia, and the European Union have proposed plans for LEO mega constellations. It is expected that approximately 100,000 constellation satellites will be launched in the next decade, far exceeding the total number of satellites launched since the beginning of human space activities. The impact of LEO mega constellations on the space environment has been revealed in recent years [3–5].

The development of LEO mega constellations poses unprecedented challenges to the long-term sustainability of space activities, as well as serious impacts on existing space surveillance and governance. A systematic study of the development, impact, surveillance, and governance approaches of LEO mega constellations is beneficial for advancing the coordinated development of the system framework and developing key technologies for the complex and integrated application of low-orbit giant constellations. Therefore, this paper introduces the latest developments in LEO mega constellations and analyzes their impacts on space activities, such as astronomical observation, in-orbit spacecraft safety, and the evolution of the space environment. The main surveillance means and governance methods for LEO objects are

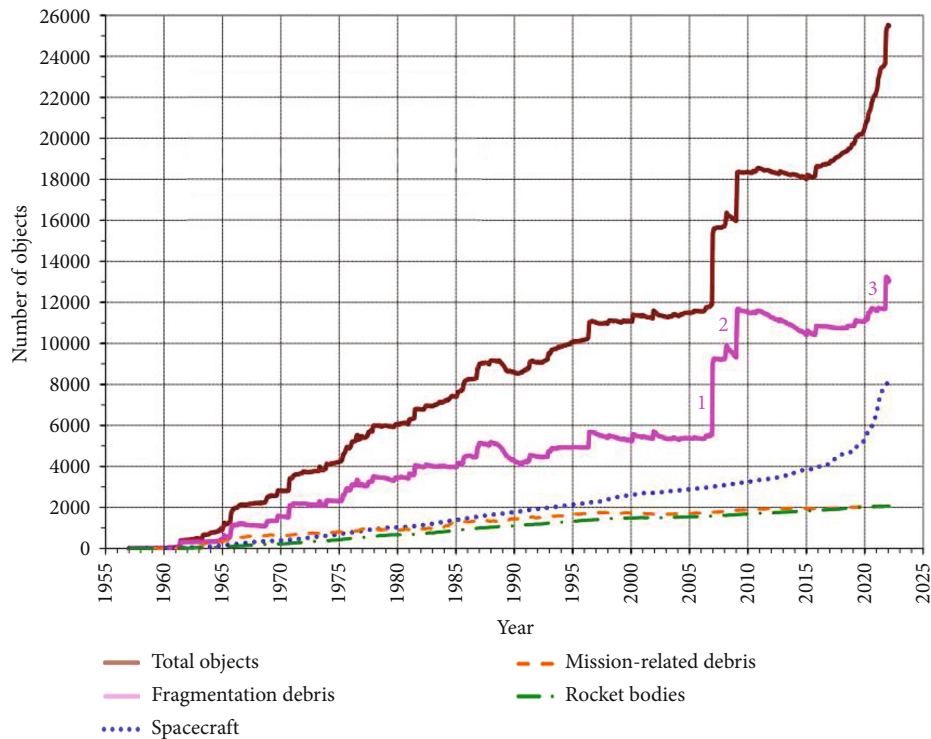


FIGURE 1: Number of cataloged space objects by NORAD [2].

introduced, and the paper concluded by the discussion of the future developments and research directions of LEO mega constellations.

2. Recent Developments

Currently, in the development of LEO mega constellations, the most representative example is SpaceX's Starlink, which plans to build a LEO constellation containing 42,000 satellites to achieve global coverage, high-speed, large-capacity, and low-latency space-based global communication system. Other well-known LEO mega constellations under construction include OneWeb, Iridium Next, Globalstar, and Flock. In addition, Samsung, Boeing, Telesat, and Amazon have proposed LEO mega constellations containing hundreds to thousands of satellites.

2.1. Starlink. Starlink is the world's largest LEO internet constellation, with a planned total of 42,000 satellites. Among them, 12,000 satellites were initially expected to be deployed in orbits around 340, 550, and 1150 km above the ground [6]. However, SpaceX has since made several modifications to its deployment plan [7] and, currently, Starlink has canceled satellites with orbital altitudes of approximately 1150 km. In 2019, the application of SpaceX for placing an additional 30,000 broadband satellites in LEO was approved by the Federal Communications Commission (FCC) [8].

Starlink satellites use lasers to communicate with each other, and satellites capable of exchanging information are adjacent in the same orbit and on adjacent orbital planes. Among the initially planned 12,000 satellites, approximately

4400 satellites operate in the Ka-band (27-40 GHz) or Ku-band (12-18 GHz), and the remaining satellites operate in the V-band (60-80 GHz) [9] (Table 1). The user terminal of Starlink adopts a phased-array antenna, and the nearest Starlink orbital plane can be aligned by the antenna aperture for the precise positioning of satellites for simple operation.

Starlink has many satellites, wide surface coverage, and extremely low network latency, which is expected to be even lower than traditional optical fiber transmission in the future [10]. Therefore, its applications are extensive. Starlink has shown excellent performance in related fields, such as international aviation and ocean transportation [11]. Moreover, the great military implications of Starlink is emerging. Its 42,000 satellites will reduce the availability of LEO frequency-orbit resources, occupying a dominant position in the LEO space environment. Furthermore, Starlink can be constructed as a powerful command and communication network. In April 2022, SpaceX sent 5,000 Starlink internet terminals to Ukraine, with the assistance of the United States Agency for International Development (USAID), providing powerful internet services during the Russian-Ukrainian conflict [12]. Starlink is an important symbol of the weaponization of outer space in the United States [13].

2.2. OneWeb. OneWeb mainly develops LEO small-satellite communications and plans to place 684 satellites in orbit [14]. These are LEO satellites with an orbital height of 1200 km, distributed on 18 orbital planes, and the included angle between adjacent orbital planes is 10.15° [15]. However, OneWeb halted its launch plans and, as of March 2022, only 428 satellites are in orbit. However, on March 21, 2022,

TABLE 1: Features of the initial 12,000 Starlink satellites.

Stage	Number of satellites	Orbit altitude (km)	Working frequency	Distinction
1	1584	550	Ku/Ka	Distributed on 24 orbital planes with an inclination of 53°. Each orbital plane plans to deploy 66 satellites.
2	2825	1110, 1130, 1275, 1325	Ku/Ka	Multiple orbital planes are deployed at each orbital altitude, with 50-75 satellites per orbital plane.
3	7518	345.6, 340.8, 335.9	V	Orbital planes distributed over three heights and inclinations.

OneWeb and SpaceX announced an agreement in which OneWeb would resume its launch schedule [16].

The working frequencies of OneWeb are the V-band, Ka-band, and Ku-band [17]. In addition, the satellites use progressive pitch technology, which can avoid interference with GEO's Ku-band satellites by slight adjustments to the pitch [18]. Simultaneously, OneWeb users are constantly replacing satellites to serve them, and the quality of communication can be guaranteed through multibeam coverage.

OneWeb has several applications and a wide range of clients. For example, in October 2021, OneWeb and Tampnet announced further cooperation [19]. OneWeb complements Tampnet's business model, especially in geographic expansion, Long-Term Evolution (LTE) backhaul, and usable backup solutions for existing clients.

2.3. Other Representative Constellations. Iridium Next is a typical LEO communication constellation, and the system plans to have 81 satellites. Currently, a total of 75 satellites have been deployed, evenly distributed on six orbital planes, of which 66 are working satellites and nine are backup satellites in orbit. The working frequencies of Iridium Next are the L-band (1-2 GHz) and Ka-band. This system has the advantages of flexible networking, multitransmitting, single-receiving, and two-way transmission [20].

Globalstar is a LEO communication system with a cumulative launch of 84 satellites. The first generation planned to launch 48 satellites and later added 12 satellites. However, owing to the rapid degradation of the S-band antenna amplifier of the satellite system, its two-way communication service was seriously affected, after which the Globalstar system was upgraded by launching 24 second-generation satellites [21]. The working frequencies are the L-band, S-band (2-4 GHz), and C-band (4-8 GHz). Globalstar has the functions of communication, maintaining nominal position and attitude, determining orbital position and velocity, and informing the ground by telemetry [22].

In addition to the abovementioned communication constellations, the remote sensing constellation, Flock of Planet Labs, also consists of hundreds of microsatellites. More than 120 nanosatellites have been launched that are small and lightweight but have an extremely high-resolution [23]. In addition, there are many LEO satellite systems still planned for launch, such as the Samsung satellite system with 4600 V-band satellites [24], Boeing satellite system with 1396-2956 V-band satellites [25], Telesat satellite system

with at least 117 Ka-band satellites [26], and Amazon's Kuiper satellite system with 3236 satellites [27].

Table 2 summarizes different characteristics of some representative LEO mega constellations. For the orbital height, although the Very Low Earth Orbit (VLEO) near 340 km can further reduce the communication delay, the satellites here bear a greater orbital maintenance cost due to the denser atmosphere. For the satellite working frequency, owing to the limited receiving ability of the antenna, satellites in the same frequency band and coverage area can only be separated by a certain angle so that the ground station can distinguish the signals of different satellites. Therefore, satellites in the same frequency band on the same orbit have certain quantitative restrictions. In addition, the signal propagation loss varies between different frequency bands. Among them, the loss of signal utilizing the range of 0.3–10 GHz radio frequency is the smallest. The Ka-band network in low-orbit experiences the largest loss. Therefore, the main working frequency bands used by LEO mega constellation plans are highly concentrated, resulting in increasingly tight satellite frequency resources [28].

3. Impact on the Space Environment

The accelerated deployment of LEO mega constellations by various countries and agencies worldwide yields a range of impacts on the space environment. From the perspective of space science, such impacts are particularly prominent in astronomical observations, in-orbit spacecraft safety, and the evolution of the space environment.

3.1. Astronomical Observations. Astronomical observations are one of the most important tools in space science [29]. According to the location, astronomical observation equipment can be categorized as ground-based (such as the Green Bay Radio Astronomy Telescope [30]) or space-based (such as the Hubble Space Telescope [31]). For ground-based devices, the field of view must traverse the atmosphere and near-Earth space to focus on the target, making these devices more vulnerable to the complex near-Earth space environment.

The new LEO mega constellations will mainly be deployed at 350-1100 km, which will significantly affect the normal operation of ground-based astronomical observation equipment. For ground-based optical telescopes, when a satellite passes through its field of view, it causes different degrees of damage to the observational data depending on the satellite brightness, as shown in Figure 2. This effect is

TABLE 2: Main LEO mega constellations' characteristics.

Constellation	Number of satellites	Orbit altitude/km	Working frequency
OneWeb	648	1200	V, Ka, Ku
Iridium Next	81	780	L, Ka
Globalstar	24	1414	L, S, C
Flock	>120	—	—
Samsung	4600	1500	V
Boeing	2956	1200	V
Telesat	>117	1000-1248	Ka
Kuper	3236	590~630	—

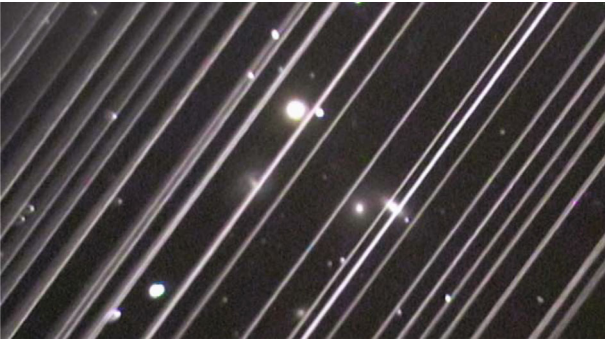


FIGURE 2: Trails from Starlink satellites passing through the field of view of a telescope during an observation, shortly after launch [34].

particularly evident in wide-field telescopes and automated patrol telescopes [32]. For ground-based radio telescopes, satellites passing through their antennas and the radio silence zone can cause the observatory to receive tens of thousands of interfering signals, which can seriously hinder observations and imaging [33].

Faced with such challenges, SpaceX proposed the testing of a satellite coating [35] and designed to reduce the reflectivity of Starlink satellites. However, reducing the brightness of satellites only has a limited effect on ground-based optical observations and does not have a positive effect on ground-based radio observations [36]. Therefore, the International Astronomical Union (IAU) [34], American Astronomical Society (AAS) [37], and National Aeronautics and Space Administration (NASA) [38] have all expressed strong concerns regarding the impact of LEO mega constellations on astronomical observations.

3.2. In-Orbit Spacecraft Safety. Ensuring the safety of in-orbit spacecraft is a prerequisite for space activities. The relative velocity between space objects can reach over 10 km/s. Therefore, potential collisions may lead to catastrophic consequences, such as spacecraft failures or even disintegration. In recent decades of human space activities, there have been several spacecraft collisions [39–41].

In recent years, the safety of in-orbit spacecraft has received increasing attention as the number of LEO space objects has grown dramatically, owing to LEO mega constellation developments. Figure 3 shows the number of close encounters between Starlink satellites and spacecraft of other

operators. In 2019, the Aeolus satellite, from the European Space Agency (ESA), increased its orbital altitude through orbital maneuvers to avoid collision with Starlink-44, from the United States [42]. In 2021, Starlink-1095 and Starlink-2305 satellites successively approached the Chinese Space Station (CSS), causing the CSS to take emergency collision avoidance measures, posing a serious threat to normal operations and the health and safety of astronauts [43]. In April 2022, OneWeb moved satellite number 0178 to avoid close contact with Starlink-1546 [4].

However, although SpaceX claims to have automatic collision avoidance systems and mechanisms [45], none of the above events demonstrated an effective response mechanism to maintain the safety of spacecraft in orbit. In fact, in a complex system containing tens of thousands of satellites, it is unrealistic to assume that propulsion systems, detection systems, and software are 100% reliable. Furthermore, for satellites with autonomous collision avoidance capabilities, if they attempt to avoid other satellites by performing a series of maneuvers, this is more likely to increase rather than decrease the risk of collision. There is no explicit space traffic management system to regulate such interactions.

Therefore, the excessive number of satellites and poor management capabilities of LEO mega constellations pose a serious threat to the safety of spacecraft in orbit. Especially for large, manned spacecraft of high value, this not only increases the risk of significant economic losses but also threatens astronaut safety.

3.3. Space Environment Evolution. In addition to posing a threat to the safety of individual spacecraft in orbit, LEO mega constellations increase the uncertainty of space environment evolution. In 1978, the Kessler effect [46] was proposed, where if the growth rate of space objects was maintained, the amount of debris would grow exponentially, owing to collision cascade effects.

Furthermore, as LEO is the most crowded region in near-Earth space [1], it has been subjected to serious accidents with long-term effects that are difficult to eradicate. In 2009, Iridium-33, from the United States, collided with Cosmos-2251, from Russia, in LEO at a relative velocity of 11.7 km/s [47]. As of July 2011, the United States Space Surveillance Network (SSN) cataloged more than 2000 pieces of space debris due to this event. Some of this debris will

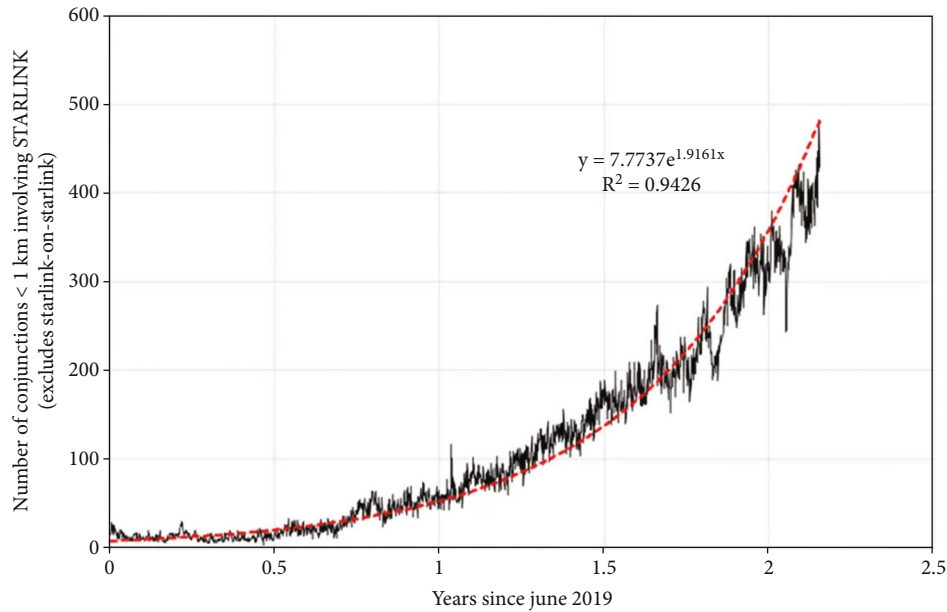


FIGURE 3: A graph showing the number of close encounters between Starlink satellites and spacecraft of other operators plotted by Professor Hugh Lewis based on data from the Socrates database [44].

remain in orbit for more than 100 years [48], potentially triggering the Kessler effect [49].

Moreover, the number of uncontrollable targets has significantly increased with LEO mega constellations, which may lead to the eventual collapse of the space environment. In February 2022, 40 of the 49 Starlink satellites launched by SpaceX were unsuccessful in initial orbit owing to the impact of geomagnetic storms [51]. Furthermore, the lifetime of satellites in LEO mega constellations is typically designed to be 3–5 years, and thousands of defunct satellites generated by such constellations may have an unpredictable impact on the space environment after the mission is complete.

The rapid growth of LEO mega constellations has led to a sharp increase in the density of LEO space objects, posing significant challenges to space debris mitigation and space traffic management. Analysis of the space environment by the Inter-Agency Space Debris Coordination Committee (IADC), NASA, and ESA, focusing on the proliferation of mega constellations [5, 52–54], highlighted that according to current trends, by 2050, the number of debris pieces larger than 10 cm will exceed 50,000, and the number of LEO satellite collisions will increase 6-fold [52]. An effective strategy to address this problem is to remove space objects, as shown in Figure 4. Therefore, to maintain the safety of future space activities and the long-term sustainability of the space environment, active space surveillance and governance must be implemented.

4. Space Target Surveillance

LEO mega constellations introduce more complexity and uncertainty to the space environment. The process of mitigating or suppressing this negative impact can be divided into two major aspects: surveillance and governance of

space objects. Toward space target surveillance, many institutions and scholars have made several research efforts [55–57] and formed an applied field of space situational awareness (SSA) with a complete architecture. Furthermore, as a means of disposition following surveillance, space target governance has been widely tested and demonstrated [58–60].

SSA [57] aims to detect, understand, and predict the exact physical locations of space objects to ensure the safety of space activities. The main data input depends on the observation equipment, and the performance of the equipment determines the number of observed targets, accuracy of the observations, and other important indicators, which are important for enhancing the capability of SSA. Given the increasing number of objects in LEO constellations, SSA has two key challenges: multisensor management and data fusion.

4.1. Observation Systems. An observation system mainly includes two deployment locations, ground-based and space-based, and two detection methods, optical and radar. Currently, the best space observation system in terms of global performance is the SSN, from the United States, followed by the Russian Space Surveillance System (SSS) and the European Union Space Surveillance and Tracking System (EUSST). To analyze the surveillance capability of observation systems for LEO mega constellations in terms of equipment characteristics, three system categories are described: ground-based radar, ground-based optical, and space-based.

4.1.1. Ground-Based Radar Systems. The United States ground-based observation system consists of more than 30 radars, optics and command-and-control centers, which can track 5 cm LEO targets and 1 m GEO targets [61]. In particular, ground-based radar systems consist of low-frequency

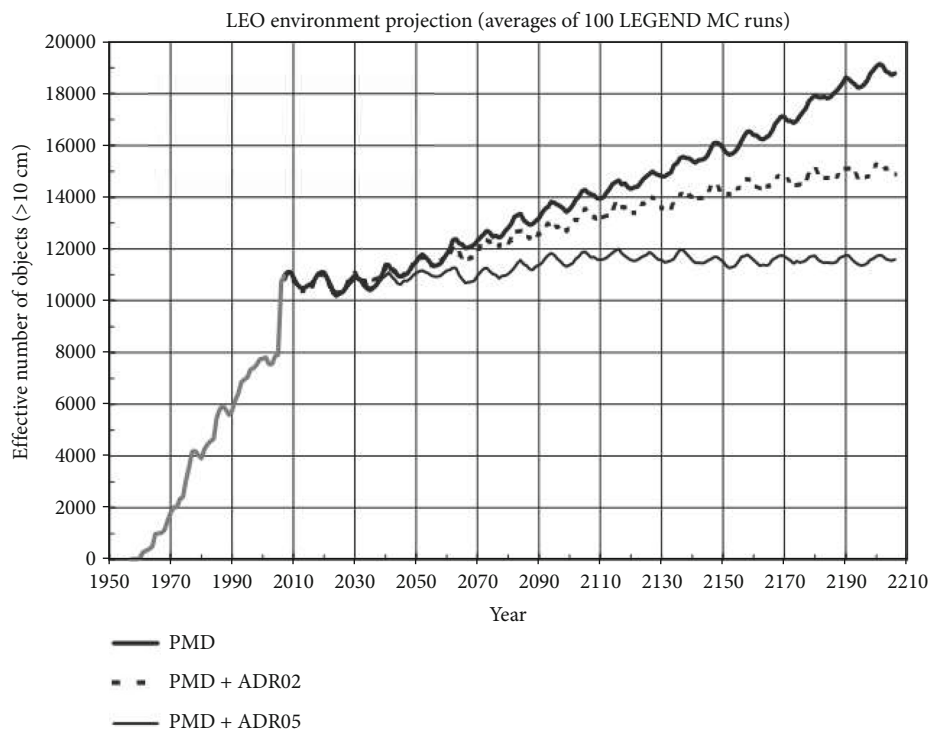


FIGURE 4: Effect of space target governance on the space environment (PMD: postmission disposal; ADR: active debris removal) [50].

radars, high-frequency radars, and combined high- and low-frequency radars that are deployed globally, for integrated detection of LEO and GEO targets. To enhance the performance of ground-based radar systems, the United States has made efforts to develop a new generation of space fence systems (SFS) in the S-band [62], shift radars in the C-band, and upgrade the original ground-based radars.

The Russian ground-based radar system mainly includes ground-based early warning radar networks and dedicated space surveillance radars in the X-band, which are mainly designed for missile early warning and LEO target surveillance over the entire territory of Russia but have almost no effect on GEO targets. In addition, SSS is mainly used for the detection and tracking of LEO targets and rarely includes identification.

Europe has less ground-based radar equipment, mainly based on Grand Réseau pour la Veille Spatiale (GRAVES), from France, and Tracking and Imaging Radar (TIRA), from Germany, as shown in Figure 5. These systems are supplemented by the French ARMOR radar and the British Chilbolton radar [63], which primarily observe LEO targets using low-frequency operating modes.

As ground-based radar systems are the main type of observation equipment in LEO, the large number of LEO targets generated by LEO mega constellations requires improvements to the precise identification capability of space objects by developing new high-frequency radars and upgrading the performance of old radars.

4.1.2. Ground-Based Optical Systems. The United States ground-based optical system includes three dedicated telescopes for Ground-Based Electro-Optical Deep Space Sur-

veillance (GEODSS), as shown in Figure 6, one Space Surveillance Telescope (SST), and five assisted telescopes for the Maui Space Surveillance Site (MSSS), primarily for observing deep-space objects. To improve performance, the United States is fully upgrading the GEODSS system and moving the SST to the southern hemisphere.

Russia's ground-based optical systems rely on the Okno system and International Scientific Optical Observation Network (ISON), which are mainly used for the detection and identification of deep-space objects. In particular, the Okno system is capable of nighttime detection of space objects at orbital altitudes of 2000-40,000 km, and ISON allows the continuous surveillance of all GEO targets.

The European ground-based optical system has a complex composition, including the ESA space debris telescope, French SPOC, German SMARTnet, and Swiss ZimLAT telescope [66], which are mainly used to observe GEO targets as small as 10 cm and some equipment is capable of night-vision observations.

In summary, for the LEO mega constellations, although the ground-based optical systems are mainly used for observing GEO targets, they are also gradually developing the capability to track fast-running LEO targets in terms of faint target observations and high-resolution imaging.

4.1.3. Space-Based Systems. Compared with ground-based observation systems, there is limited experience with space-based observation systems. The United States has developed several space-based projects for GEO and LEO, including Space-Based Space Surveillance (SBSS) [67] (Figure 7), STSS, ORS-5, and GSSAP, and has shared data with Canadian Sapphire satellites [68] to enhance the surveillance capability



FIGURE 5: The Tracking and Imaging Radar (TIRA) [64].



FIGURE 6: Ground-based Electro-Optical Deep Space Surveillance (GEODSS) [65].

of GEO targets and imaging of deep-space objects. In particular, Block-10 in SBSS can operate under all weather conditions, and its revisit capability far exceeds that of optical telescopes on the ground with a larger aperture, which shortens the cataloging update cycle of GEO targets to 1-2 days, demonstrating the unique advantages of space-based observation systems.

Complementing existing ground-based observation equipment, space-based observation systems have the advantages of high-precision orbit determination and high-resolution imaging because they overcome geographical constraints and atmospheric influences. ESA plans to launch

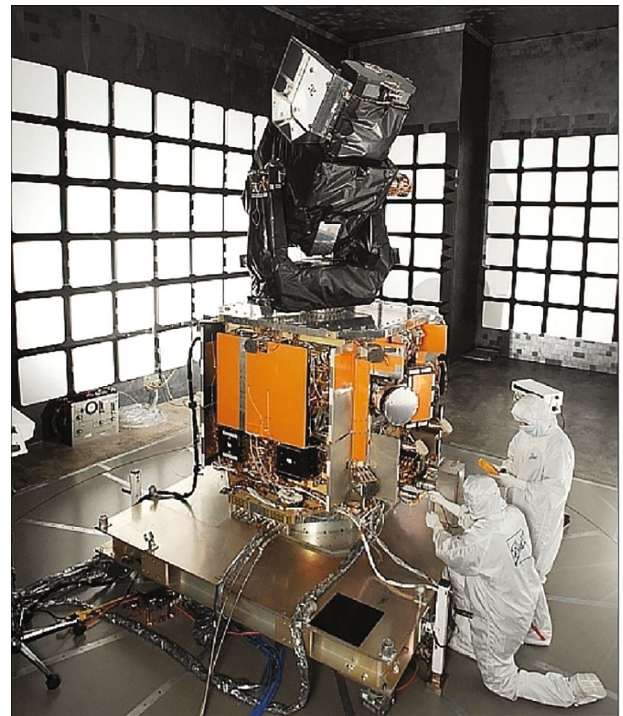


FIGURE 7: SBSS-1 pathfinder spacecraft during integration at BATC [69].

space-based observation equipment in 2025 to detect space objects as small as several millimeter in diameter [70]. In addition, the fast revisit capability of space-based equipment has greater potential in meeting the observation challenges of LEO mega constellations.

4.2. Space Situational Awareness. Currently, SSA is facing new challenges in terms of multisensor management and data fusion, owing to the development of LEO mega constellations. As of March, 2022, the number of cataloged space objects is around 26,000, whereas Starlink has proposed a

plan containing 42,000 satellites, which is far beyond the current capacity of the catalog. It is difficult to complete efficient real-time surveillance with only a limited number of sensors when facing such a large number of space objects. Therefore, to maximize the capabilities of SSA, an efficient allocation of multisensors is required, with an effective fusion of multisensor data.

4.2.1. Multisensor Management. In SSA, the multisensor management method can be understood in terms of sensor scheduling or the dispatch of observation tasks, which refers to the allocation of appropriate observation instructions at appropriate times, so that the entire sensor network can work together to achieve task requirements. With the increasing number of ground-based and space-based observation sensors coming online, effective multisensor management methods become an urgent demand by the space community.

Traditional sensor management methods in SSA can be broadly divided into heuristic and information-theoretical methods. Heuristic methods generate sensor tasking solutions without involving state information and mainly focus on a few evaluation indicators, such as improving the signal-to-noise ratio of measurement data or reducing the uncertainty of cataloging targets [71, 72]. The information-theoretic method primarily determines the observation order of the target by maximizing the information gain of the control instruction. As greater information gain yields more significant reduction of the posterior uncertainty, it can be expected that the observation instruction with a larger information gain can bring about a more accurate state estimation.

Using the principle of maximizing the Cauchy-Schwarz or Rényi information gain, Gehly et al. [73] designed a control method applied to the GEO cataloging maintenance task. Cai et al. [74] derived the Rényi information gain expression for an LMB random finite set and designed a fast solution algorithm for suboptimal control instructions. Combining heuristic methods with information theory is a new trend. Harris et al. [75] proposed a space-based sensor swarm design that combined information-based sensor tasks and heuristic optimization. In large and complex space-based sensor design, the use of heuristic optimization methods provides a balance between local search and stochastic exploration, which is beneficial for traversing complicated nonconvex solution spaces.

The SSN is the most representative SSA system. Currently, SSN mission planning relies on the Space Defense Operations Center (SPADOC), which employs a special perturbation (SP) mission system to help maintain the accuracy of the catalog. The SP tasker provides an approximate daily schedule to the SSN, which in turn provides information in the form of sensor observations to help refresh the two-line element (TLE) set. The United States Space Force (USSF) plans to use a sensor network autonomous resilient extensible (SNARE) system to decentralize the tasking and collection of SSN sensors [76]. SNARE turns the SP tasker's catalog-keeping capabilities into a real-time stream of location data, allowing each sensor to read trusted, decentralized

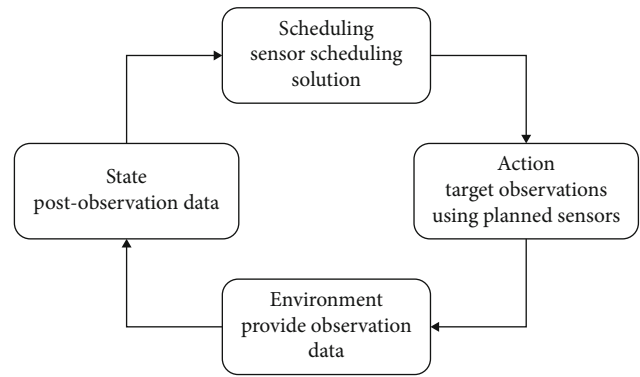


FIGURE 8: Multisensor management.

information for individual processing. This distributed management method can improve the timeliness of the positioning information and positioning accuracy. Multisensor management as shown in Figure 8.

Currently, multisensor management suffers from the “curse of dimensionality” problem; that is, as the number of targets and the length of the observation window grow linearly, the complexity of the management problem increases exponentially. In addition to typical optimization methods, Roberts et al. [77], Faber [78], and Linares and Furfaro [79] have proposed efficient and optimal task allocation methods based on deep reinforcement learning algorithms and related methods to achieve good performance in high-dimensional and large-scale scenarios.

4.2.2. Multisource Information Fusion. Multisource information fusion is a multilevel and multifaceted process of information processing that detects, correlates, and combines data from multiple sensors and information sources to obtain an accurate estimate of the target status and identity, as well as a complete assessment of environmental posture and threats. The functional model of multisensor information fusion is shown in Figure 9 [80]. The main theories and methods include the voting method [81], clustering method [82], Bayesian inference [83], Dempster-Shafer theory [84], and neural networks [85].

In practical applications, the United States uses ground-based and space-based sensors as the starting point of information fusion for SSA and transmits the acquired data through the SSN to the three major command-and-control centers, including the Space Surveillance Center (SSC), the Combined Space Operations Center (CSPOC), and the National Space Defense Center (NSDC), further supporting the Joint Force Space Component Commander (JFSCC) to command joint operations in each theater. However, Europe's EUSST integrates sensors distributed across member states into a unified sensor network that operates, controls, and correlates sensors based on user-generated requirements and transmits the collected SSA data to the European database in real-time.

In addition to the national level, many universities and commercial organizations have deployed SSA systems [86–88]. For example, the Optical tracking and Spectral characterization of CubeSats for Operational Missions

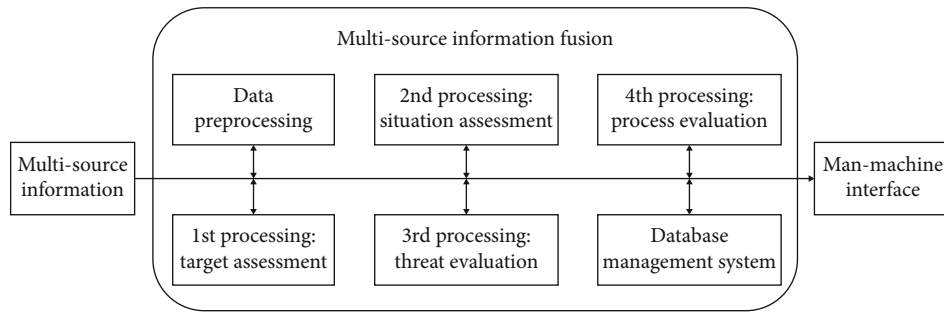


FIGURE 9: Functional model of multisource information fusion [80].

(OSCOM) [87] system from Embry-Riddle Aeronautical University costs only \$10 K and is capable of producing high rate photometric measurements of LEO CubeSats to estimate spin rates. The University of Texas at Austin has established ASTRIANet [89], a global network of ground-based astronomical telescopes, to support SSA studies such as orbit determination and information fusion by collecting optoelectronic observations. Numerous related projects have demonstrated the positive role that low-cost, small-aperture telescopes can play in SSA systems [90]. In addition, a commercial company LeoLabs has established a global phased-array radar network [91], including two ultra-high frequency (UHF) sites and two S-band sites, which detect nearly 20,000 targets in a single day and can provide ephemerides and conjunction data messages for many LEO targets in real-time. Furthermore, to achieve the integration of space surveillance networks and data, the United States Defense Advanced Research Projects Agency (DARPA) proposed the OrbitOutlook program in 2015, which initially realized the capability to process multisensor data provided by the government, military, civil, commercial, scientific, and international partners and provides higher quality space environment surveillance and collision risk prediction through information fusion.

However, current multisensor information fusion experiences limitations, such as low autonomy and poor timeliness. The development of LEO mega constellations increases the complexity and uncertainty of information, posing new challenges to SSA and space surveillance. Therefore, developments in the autonomy, rapidity, and effectiveness of multisensor information fusion are urgently required.

5. Governance Methods

To mitigate the damage caused by LEO space objects that become trapped in orbit and protect the space environment, the IADC “Space Debris Mitigation Guidelines” were issued in 2002 [92]. They stipulate that the lifetime of a LEO space target following disposal should not exceed 25 years. Currently, there are two main governance methods. The first category, postmission disposal (PMD), is to reduce the generation of new space objects by onboard deorbiting strategies. The second category, active debris removal (ADR), mainly aims to speed up the deorbit of out-of-service space objects, and the ultimate goal is to crashing targets into the atmosphere through active human activity.

5.1. Postmission Disposal. The International Academy of Astronautics (IAA) classifies postmission deorbit devices into the following four categories: combustion propulsion systems, drag augmentation, electrodynamic tether (EDT), and solar sail deorbit [93], as shown in Figure 10.

5.1.1. PMD Devices. Propulsion deorbit devices obtain thrust by consuming fuel to directly deorbit the spacecraft. Drag augmentation deorbit devices use atmospheric drag to decelerate, thereby lowering the orbital altitude of the spacecraft, which mainly includes off-orbit sails and drag balls. EDT devices rely on the electromagnetic environment surrounding the Earth for the conductive tether to interact with the geomagnetic field and generate the Lorentz force, which is used to change the orbit of the spacecraft. Solar sails rely on reflected solar radiation to generate light pressure thrust, which forces the spacecraft to leave its original orbit.

The propulsion deorbiting device is suitable for various track heights, stable, and reliable. For example, Starlink uses electric propulsion method to deorbit after a mission. Using the propulsion system to deorbit the spacecraft is efficient; however, this action comes at a high cost because the life of the spacecraft is related to the amount of fuel carried. For the drag enhancement device, because the atmospheric density is closely related to the track height, the efficiency of the drag-enhancing device in lowering the orbit of a spacecraft increases with decreasing orbital altitude. When the track height is greater than 1000 km, the deorbit efficiency of the drag enhancement device is significantly reduced because of the very thin atmosphere. However, the electric power rope that generates thrust by cutting the magnetic induction lines can better meet the deorbit requirements of the spacecraft within this orbit. The solar sail is more suitable for GEO spacecraft because of the weak solar-light pressure of LEO, which leads to a low deorbit efficiency of the solar sail.

5.1.2. PMD for LEO Mega Constellations. In summary, PMD methods suitable for deorbiting low-orbit giant constellations use drag augmentation or EDT deorbit devices. Off-orbit sails rely on aerodynamic resistance to descend the orbit and tend to be nonplanar in configuration design to enhance their aerodynamic stability and improve resistance efficiency. To ensure a large windward area, deorbit sails generally require spacecraft attitude control [94]. The NanoSail-D2 satellite, from the United States [95]; the

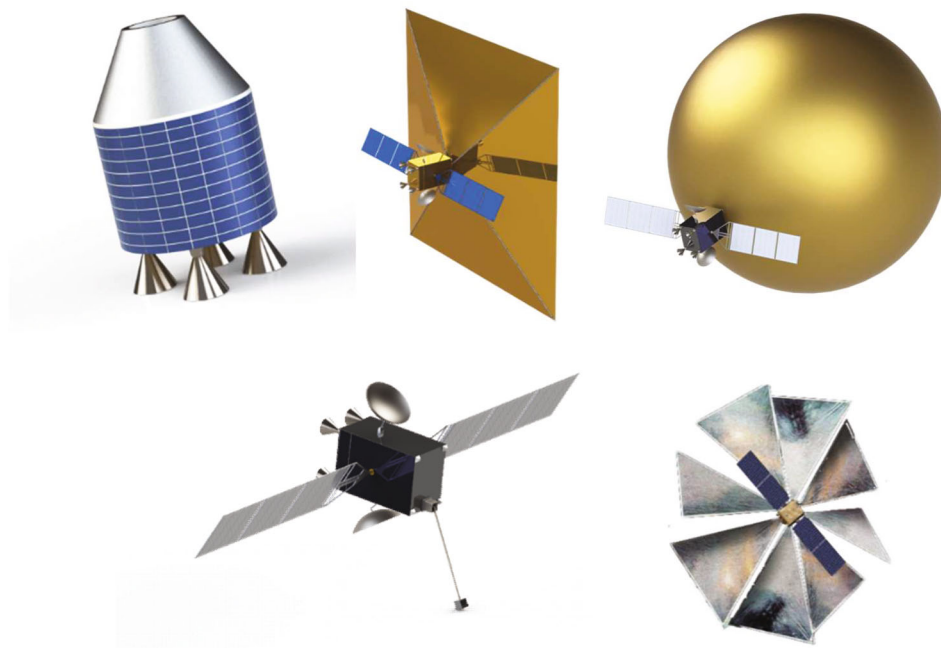


FIGURE 10: Postmission deorbit devices. (a) Combustion propulsion system. (b) Off-orbit sails. (c) Drag balls. (d) EDT. (e) Solar sails.

Inflate Sail satellite, from the United Kingdom [96]; and the QingTeng satellite, from China [97], have all conducted verification experiments for the on-orbit deployment of off-orbit sails. Currently, on-orbit deployment with a sail surface area of 10 m^2 has been achieved; however, because the efficiency of the off-orbit sail is significantly affected by the sail surface attitude, a self-stabilizing method for the sail surface has yet to be developed.

Compared with the deorbit sail method, the configuration of the drag ball exhibits omnidirectional resistance. Therefore, it is more suitable for attitude instability targets or end-of-life spacecraft where the actuator capability cannot satisfy the attitude stabilization requirements. In 2019, the Beijing Institute of Technology launched CubeSat: BP-1B [98], which has a mass of 3 kg and is equipped with a sail ball with a diameter of 0.5 m. BP-1B is the first CubeSat in the world to demonstrate the drag ball deorbit method in-orbit. To promote the application of drag balls, further research and on-orbit validation are required to determine the impact of extreme space weather and maintain long-term effectiveness when operating at higher orbital altitudes. Potential research avenues include low-damage and high-density folding and unfolding schemes for developing drag balls with super-large diameters and efficient methods for air supplementation and shape maintenance.

EDT deorbit devices have large flexibility and size and strong nonlinearity. They are affected by many physical field factors, such as the geomagnetic field, Earth's gravitational field, atmospheric resistance, and space plasma density. Various countries have carried out on-orbit tests for key technologies with different lengths of EDTs, such as on-orbit release and electronic transceivers. Furthermore, attitude stability controls, such as TSS-1 [99], STARS-1 [100], and KITE, have been applied with EDT release lengths varying from 5 m to 35 km; however, most missions failed due to

technical failures. Therefore, avoiding kilometer-level tether failures during release, deployment, and operation is a key issue in the development of EDTs, as shown in Figure 11.

Currently, most deorbit methods are still in the stage of on-orbit verification, with several theoretical and practical problems to be solved. For example, the coupling effect of forces, heat, flows, and solid of deorbiting devices is unneglectable in complex space environments. In addition, the mechanisms of many dynamics and control problems in the deorbiting process still need further investigation.

5.2. Active Debris Removal. Compared with PMD methods, active space object removal methods enable remedial governance of on-orbit spacecraft without deorbiting devices. The main methods with more mature technologies or on-orbit verification tests include laser removal, spaceborne harpoons, spaceborne nets, and space robots.

5.2.1. ADR Methods. The laser removal method [101] uses an intense laser beam to irradiate the surface of the target, causing melting, vaporization, and ionization of the material in the irradiated area, which creates a plasma expansion plume. The impulse of the expansion plume causes the target to acquire a reverse velocity increment and deorbit. The spaceborne harpoon [102] penetrates and immobilizes the target by launching a rod-like mechanism with a recovery device, thereby dragging it out of orbit. Spaceborne nets [103] capture and deorbit the target by launching an interceptor net. Space robots [104, 105] are more flexible, with multijoint mechanisms that allow them to not only deorbit and remove targets but also repair mission-specific targets, as shown in Figure 12.

The laser removal method is promising because of its short response time, reusability, and low-cost, as shown in Figure 13. The directional deorbiting characteristics of

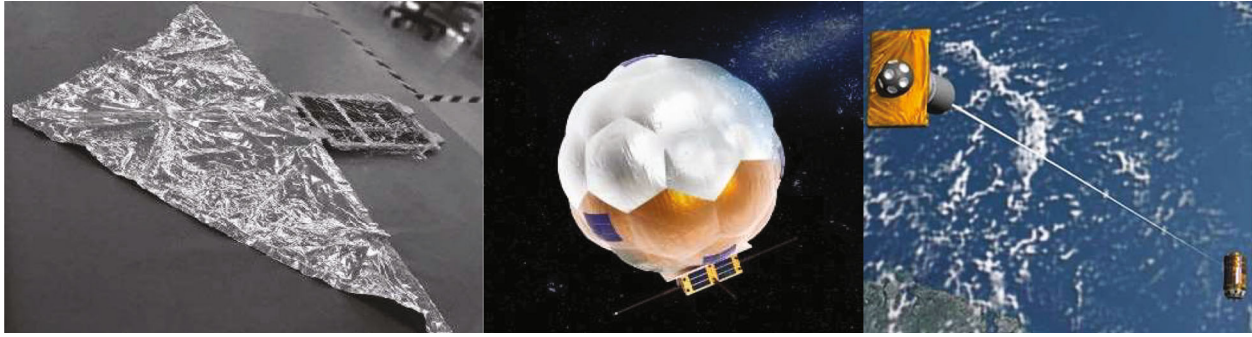


FIGURE 11: Postmission deorbit devices. (a) QingTeng. (b) BP-1B [98]. (c) KITE.

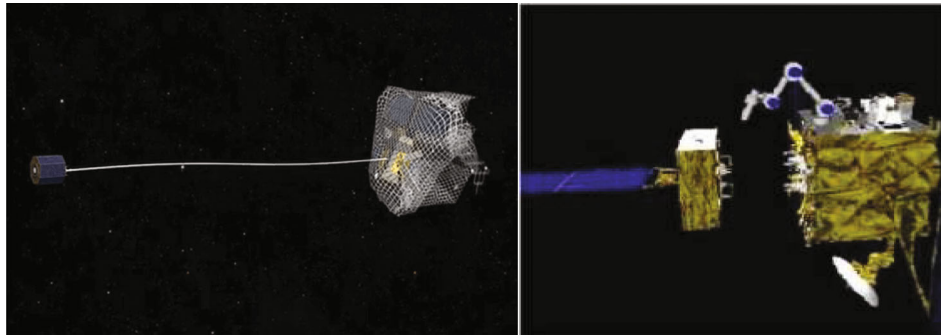


FIGURE 12: Active debris removal devices. (a) Spaceborne net. (b) Space robot.

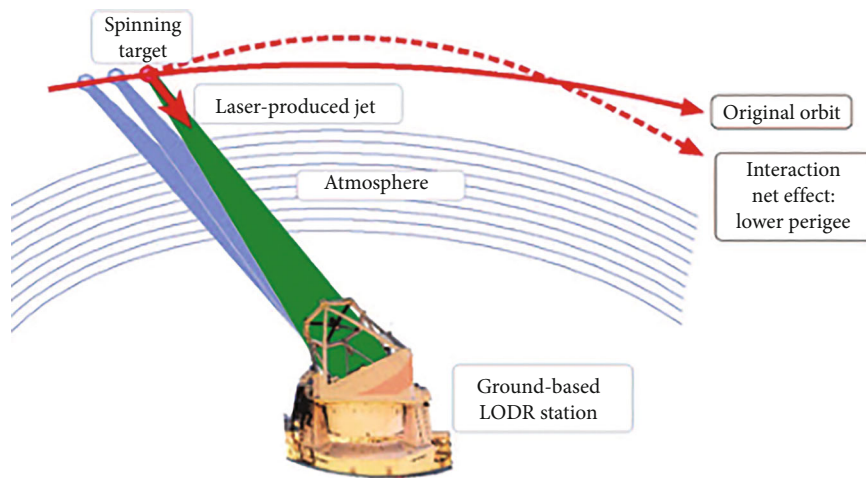


FIGURE 13: Laser space-debris cleaning [111].

spaceborne harpoons make them suitable for the removal of specific targets. For spaceborne nets, the flexible net is particularly suitable for the safe capture of a large tumbling spacecraft. The higher cost of space robots makes them more suitable for high-value mission scenarios until engineering advancements allow them to be produced at a high volume.

5.2.2. ADR for LEO Mega Constellations. In summary, the main active space target removal methods applicable to the deorbiting of LEO mega constellations are laser removal, spaceborne harpoons, and spaceborne nets. NASA proposed a ground-based ORION program in 1996 [106, 107], which

aimed to cover space objects with masses below 100 kg within 1500 km. Since 2012, owing to atmospheric obstructions due to the laser transmission problem, NASA has shifted to a space-based laser program. The Australian Space Environment Research Centre (SERC) has also proposed a ground-based laser removal program, but technical constraints have prevented the target from being reoriented as scheduled. Currently, this technology has entered key development [108, 109], system demonstration, and verification test phases [110].

Recently, the RemoveDEBRIS project, led by Surrey University, has conducted in-orbit experiments on harpoons

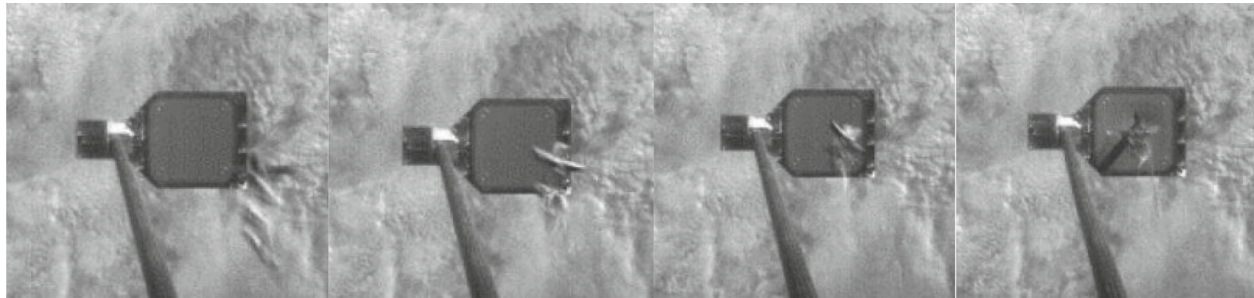


FIGURE 14: Harpoon impacting the target [112].

and nets [112]. In 2018, the net experiment succeeded. The satellite started by releasing a small CubeSat and then launched an interceptor net 6 m away from the satellite to capture it. In 2019, the harpoon test was successful in capturing a target. A piece of aluminum, simulating space junk, was attached to a carbon fiber rod extending from the satellite. The satellite aimed a pen-sized harpoon at the target, successfully penetrating the aluminum sheet and dragging it back to the satellite, as shown in Figure 14. RemoveDEBRIS was the first mission to successfully demonstrate ADR technology in orbit.

Currently, the ADR methods described above have only been validated for a single target, which should be improved in terms of removal efficiency. In addition, the removed objects were all cooperative targets, and the problem of dynamic control was considerably simplified. Therefore, there is still a research gap in applications for noncooperative targets. In addition, Susanne et al. [113] proposed the ADReS-A approach for the batch removal of spatial targets. Wang et al. [114] analyzed the use of tethered nets to capture space debris clouds. Lucken et al. [115] proposed the use of a constellation to collect space debris. Each CubeSat in this constellation can rendezvous with 50 pieces of debris within two years, avoiding expensive orbital maneuvers for frequent inclination change. In summary, there is an urgent need for further research on noncooperative multitarget bulk removal for LEO mega constellations. The design of a “debris removal constellation” to remove defunct satellites from a mega constellation is a potential solution, where the major difficulties lie in the design of its configuration and a suitable rendezvous strategy, such that making the removal more efficient.

In summary, the LEO congestion caused by the rapid growth of LEO mega constellations poses new challenges for the efficiency and economy of the existing governance methods. PMD can significantly reduce birth rates and increase the rate of space failure targets. However, this cannot curb the growth trend. ADR can dispose of existing failure targets and fundamentally curb the tendency for space junk growth. However, there is an urgent need to improve removal efficiency. Therefore, the integrated use of both PMD and the active removal of space objects is a prerequisite for ensuring the sustainability of the space environment. Specifically, the deorbiting of defunct space objects should be accelerated by developing standardized, modular, efficient, and engineered means of governance.

6. Conclusion

6.1. Future Development Trends. Comprehensive applications of LEO mega constellations are still in the stage of preliminary exploration due to some unique characteristics, such as limited frequency-orbit resources, global impact, and complex constraints. The main future development trends are discussed in the following sections.

6.1.1. Bellwether Firms Rapidly Reserve Frequency-Orbit Resources in Batches. To curb the uncontrolled occupation of frequency-orbit resources by LEO mega constellation plans, the International Telecommunication Union (ITU) has established a “milestone” policy [116], which requires declarants to launch a specified number of satellites within a period. Consequently, bellwether firms with first-mover advantages, such as SpaceX and OneWeb, have adopted a multiplan and multiround rolling declaration strategy to reserve frequency-orbit resources. Meanwhile, these firms internally implemented centralized and unified management of frequency-orbit resources. In terms of specific frequencies and orbits, the main frequency bands used were V, Ka, and Ku, supplemented by L, S, and C. The main orbits were LEO and VLEO, with LEO used to provide comprehensive coverage of the major human activity areas, and VLEO used to further enhance the constellation capacity and reduce the communication time delay. Moreover, because of the lower orbital altitude, it is more convenient for the satellites to accelerate the deorbiting of satellites at end-of-life.

6.1.2. Unprecedented Damage to LEO Environment. Damage to the space environment from LEO mega constellations is focused on three main areas: astronomical observations, in-orbit spacecraft safety, and space environment evolution. For astronomical observations, it is becoming increasingly difficult to obtain valid data. Images from ground-based optical observations contain light spots, and signals from ground-based radio observations are scrambled. The collision risk of in-orbit spacecraft increases with the increase in the number of satellites produced by LEO mega constellations. In particular, for large, manned spacecraft of high value, such as the International Space Station (ISS), each collision avoidance maneuver consumes approximately 30 kg of propellant. Furthermore, this results in high economic losses and threatens the lives of astronauts during maneuver. As for the space environment evolution, owing to the large

number of satellites in LEO mega constellations, uncontrollable targets resulting from accidental failures and postmission invalidation will likely direct the long-term evolution of the LEO space environment toward eventual collapse. At that stage, not only the launch windows for spacecraft resupply and rapid rescue missions can be greatly restricted but the space assets accumulated by humans over several decades will be lost.

6.1.3. Surveillance Systems Evolve from Ground-Based to Space-Based. The complexity and uncertainty of the space environment induced by LEO mega constellations require SSA systems to be equipped with stronger real-time surveillance capabilities to capture more space state changes promptly, avoiding serious accidents as far as possible. In this regard, compared with traditional ground-based radar systems and optical systems, space-based surveillance systems can overcome the limitations of geographical location and the atmosphere to obtain more effective information about the target, which is an effective supplement for surveilling massive LEO targets. Furthermore, because of its higher revisit frequency, space-based surveillance systems are capable of acquiring denser target information using multiple collaborated sensors. By further developing multisensor management and information fusion, space-based surveillance systems can potentially provide real-time surveillance, which is more suitable for meeting the challenges posed by LEO mega constellations.

6.1.4. Governance Methods Evolve from Single-Objective Targets to Multiobjective, Low-Cost, and High-Efficiency Targets. Currently, although PMD methods, such as drag augmentation and EDTs, can significantly reduce the creation of new defunct satellites, they cannot curb the growth of defunct space objects. While ADR methods, such as laser removal, spaceborne harpoons, and spaceborne nets, are effective for the governance of existing space failure targets, they are limited to one-to-one removal. For LEO mega constellations, low-cost, high-efficiency, bulk governance of multiple targets will be a high-demand in future. Such governance includes both the disposal of existing targets and the reduction in the birth of new targets.

6.2. Future Research Directions. As LEO mega constellations are characterized by many satellites, wide surface coverage, and large scale, the rules of their development and impacts on the space environment are very complex, which poses new challenges to the real-time nature of the SSS and the efficiency of the governance methods. Considering the coupling of the LEO system, the following future research directions are suggested.

6.2.1. International Frequency-Orbit Resource Allocation Protocol. Currently, the main principle for using frequency-orbit resources is “first-come, first-served” in the international arena, which causes LEO mega constellations with first-mover advantage to present a significant barrier to the development of other planned constellations. Therefore, from the bottom system, it is necessary to study a more equitable coordination framework for LEO frequency-orbit resource allocation, such

as the fair use principle proposed by ITU for GEO, which safeguards the development rights and opportunities of late-developing countries in the field of LEO constellations by allocating resources to each country. Furthermore, studies on anomaly spectrum detection, compatibility index, and shared spectrums would provide technical support.

6.2.2. Technical Standards and Disposal Mechanisms for Space Traffic Management. As an increasing number of spacecraft enter near-Earth space, space traffic becomes normalized, as with the land, sea, and air traffic, accelerated by the development of LEO mega constellations. However, due to the lack of political consensus, absence of economic feasibility, and high military sensitivity, the international community is currently divided on the issue of constructing a space traffic management system. For example, the United States and Russia have long debated the collision event of Iridium-33 and Cosmos-2251 in 2009 but neither could provide sufficient evidence to judge the disposal standards of the accident [117]. Nevertheless, the critical demands brought by increasingly severe space congestion will eventually promote dialogue and cooperation among the major parties. Until this point, a reasonable and unified technical standard for space traffic management should be established to further improve the ex-ante negotiation mechanisms, crisis control mechanisms during accidents, and ex-post accident disposal mechanisms.

6.2.3. Critical Technologies for Timely Surveillance. For the upcoming large number of space objects brought by LEO mega constellations, it is more difficult to achieve real-time surveillance using the existing ground-based system. Therefore, there is a demand to expand the surveillance network by developing space-based systems to provide more flexible observations and frequent revisit capabilities. Simultaneously, multiple sensors in the system provide multisource information in an integrated manner and, eventually, different types of information will be efficiently fused and processed. Distributed multisensor management is a promising key technology for real-time surveillance, owing to its ability to enable each sensor in the system to obtain credible information independently, which provides improved real-time surveillance accuracy. As for processing different types of information efficiently, the technology for multisensor information fusion combined with intelligent algorithms provides the advantage of easier access to consistent interpretation by machines in massive heterogeneous information, which significantly improves the processing efficiency of multisensor data and the understanding of information characteristics, thus achieving real-time surveillance of a large number of LEO space objects.

6.2.4. Key Methods for Efficient Governance. Efficient and effective governance methods are urgent demand to handle the potential large number of defunct satellites in LEO mega constellations. For thousands of satellites in normal operation, further research on multitarget, low-cost, bulk deorbiting strategies should ensure that massive targets in postmission invalidation are safely removed from orbit and purified by falling into the

atmosphere. However, for satellites in orbit with abnormal damage, there is an essential demand for intelligent and rapid response governance technology in complex environments. Through information-driven autonomous decision-making and rapid response capabilities, out-of-service targets are stripped from busy space traffic networks promptly, avoiding far-reaching impacts to space environment.

Data Availability

The data of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Authors' Contributions

J. Zhang and H. Cai conceived the idea of this review and supervised the study. Y. Cai and C. Xue wrote the manuscript. H. Cai and Z. Xue revised the manuscript. All authors discussed the results and contributed to the final version of the manuscript.

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